

Exploratory Data Analysis: E-Commerce Orders

Project 2 | DecodeLabs Data Analytics Internship | 20 September 2026

1. Problem Statement

The business holds 1,200 order records spanning January 2023 to June 2025 but has no view of what drives revenue, whether sales are growing or declining, or whether any transactions warrant investigation. This analysis interrogates the dataset to answer three questions: **What is the shape of the data? Are there real trends or outliers? And which findings should actually change a business decision?**

2. Methodology

Analysis was performed in Python (pandas, NumPy, SciPy, Matplotlib, seaborn) via a single reproducible script, `eda_analysis.py`. Blank CouponCode entries were treated as 'no coupon applied' rather than missing data and labelled explicitly; no rows were deleted. Outliers were tested with **both** the IQR ($1.5\times$) and Z-score ($|z|>3$) methods so the two could be compared, and every flagged value was then validated against the line-total identity ($\text{TotalPrice} = \text{Quantity} \times \text{UnitPrice}$) to separate genuine errors from genuine large orders. Trend strength was judged by r^2 , not by the visual slope, and the seasonality test was corrected for unequal year coverage before being run.

3. Key Findings

Finding 1 — Order value is right-skewed; report the median, not the mean

Mean order value is **1,053.97** but the median is **823.62** (skew = 0.89). The mean is pulled upward by a tail of large orders and overstates the typical transaction by roughly 28%. UnitPrice, Quantity and ItemsInCart are all symmetric (skew ≈ 0) — the skew enters only when quantity and price multiply together.

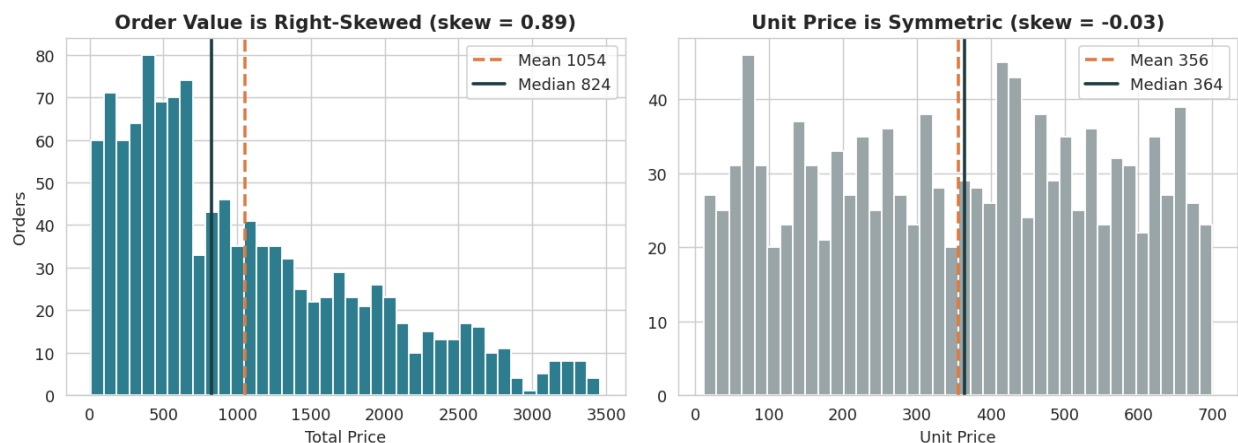


Figure 1 — TotalPrice carries a right tail; UnitPrice does not.

Finding 2 — The apparent H1 sales spike is a reporting artifact, not seasonality

Raw monthly totals show 741 orders in January–June against 480 in July–December, which reads as a strong first-half concentration. It is not. **The dataset ends on 30 June 2025**, so Jan–Jun is observed across three years while Jul–Dec is observed across only two. Normalising by the number of observed years gives **exactly 40.0 orders per month in both halves**. A chi-square test on the complete years (2023–2024) returns **p = 0.744** — no significant deviation from uniform. Acting on the raw figures would have meant shifting budget into

a half-year that has no genuine advantage.

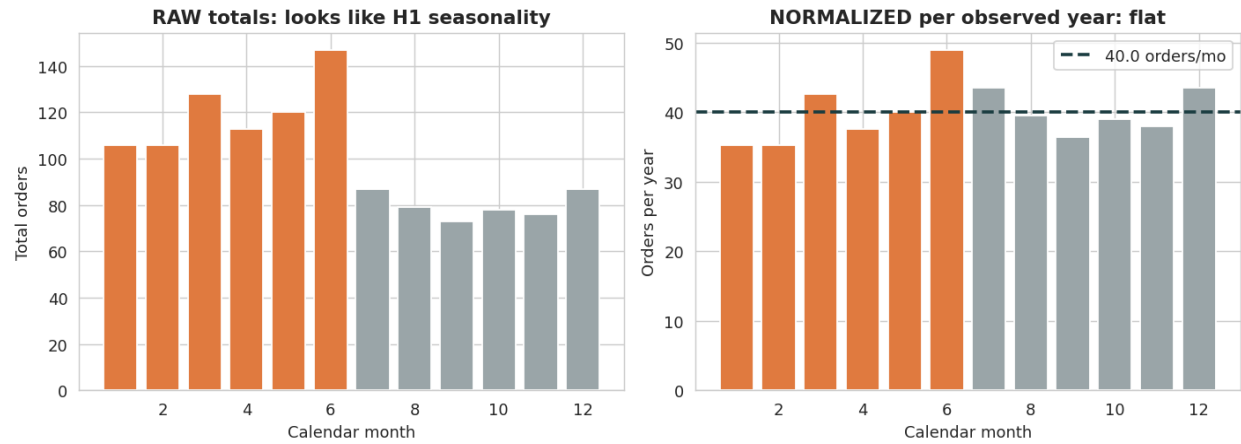


Figure 2 — The same data before and after correcting for unequal year coverage.

Finding 3 — Revenue shows no reliable trend

Monthly revenue carries a negative slope of -349 per month, but the correlation with time is only $r = -0.297$, giving $r^2 = 0.088$. Time explains under 9% of the variance in monthly revenue; the remaining 91% is noise. The year-on-year table appears to show decline, but 2025 contains six months against twelve in prior years — comparing them directly is invalid.

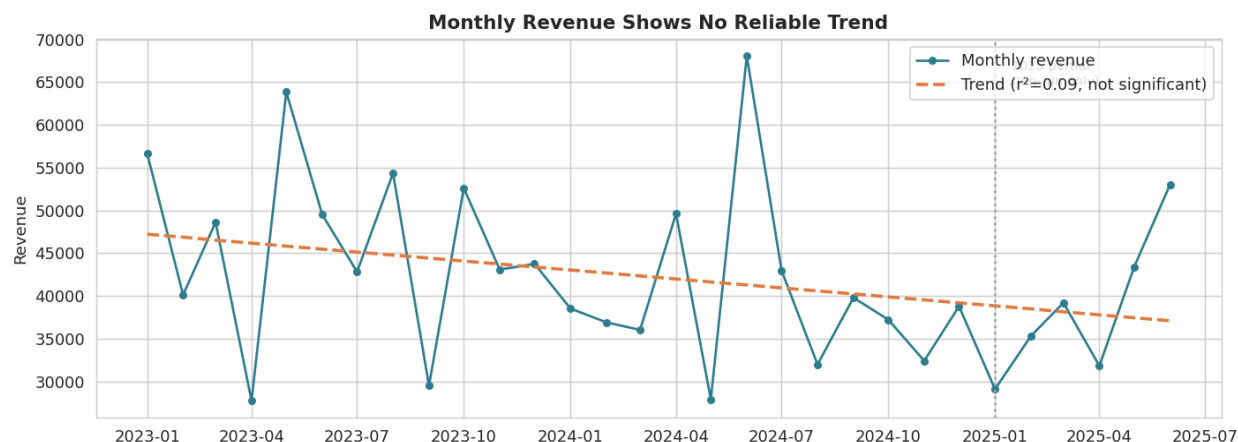


Figure 3 — Monthly revenue with fitted trend line.

Finding 4 — The 8 flagged outliers are signal, not noise

The two detection methods disagree, which is itself informative. **IQR flags 8 orders** in TotalPrice (0.67% of records, above a fence of 3,330.41); **Z-score flags none**, because the maximum $|z|$ is 2.93 — just under the threshold, and the standard deviation is inflated by the very values being tested. On bounded business data the robust IQR method is the more trustworthy of the two.

Critically, all 8 are legitimate. Every one is a maximum-quantity order (5 units) of a premium-priced item; **all 1,200 rows satisfy TotalPrice = Quantity × UnitPrice with zero exceptions**, and the largest observed order (3,456.40) sits below the theoretical ceiling of 3,499.65. These are the business's most valuable customers and should be investigated for retention — not cleaned away.

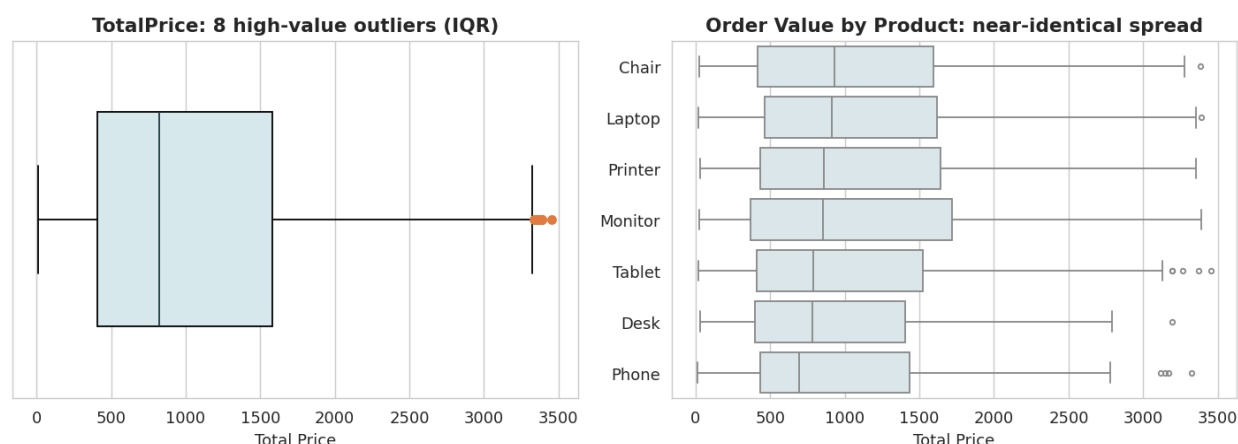


Figure 4 — Boxplot of TotalPrice, and order value by product category.

Finding 5 — Unit price drives revenue more than quantity does

UnitPrice correlates with TotalPrice at $r = 0.717$, ahead of Quantity at $r = 0.615$. Quantity and ItemsInCart move together ($r = 0.650$), as expected. But **UnitPrice and Quantity are uncorrelated ($r = 0.015$)** — customers

buying expensive items do not buy fewer of them. Note that TotalPrice is derived from the other two columns, so these correlations are partly structural; the informative result is the near-zero price/quantity relationship.

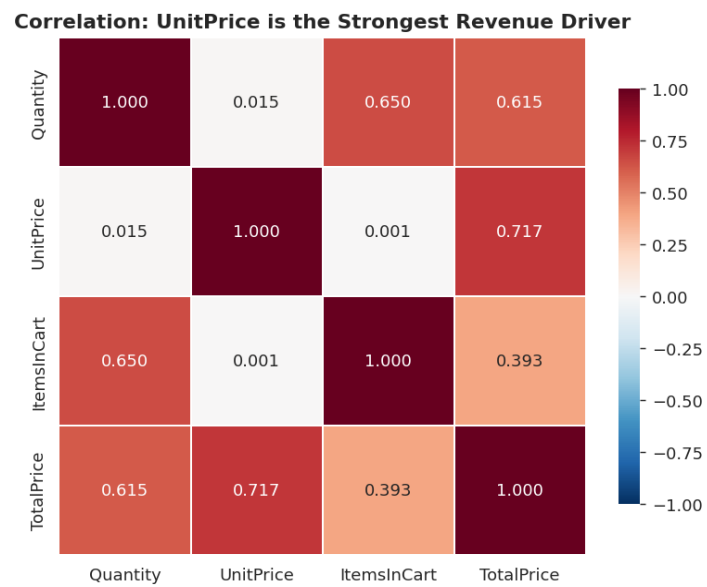


Figure 5 — Pearson correlation matrix.

Finding 6 — Every categorical variable is near-uniform

No product, payment method, referral channel or order status holds a meaningful lead. Product share spans only 13.0%–15.1% across seven categories; payment methods span 19.2%–21.5% across five. Most striking is order status: **Cancelled (20.8%) and Returned (20.6%) are the two largest categories**, while Delivered is the smallest at 19.2%. A genuine business would not show a 41% combined cancel-and-return rate spread this evenly — this flatness across every dimension indicates **synthetically generated data**, and it is the reason no segment-level insight can be drawn from this dataset.

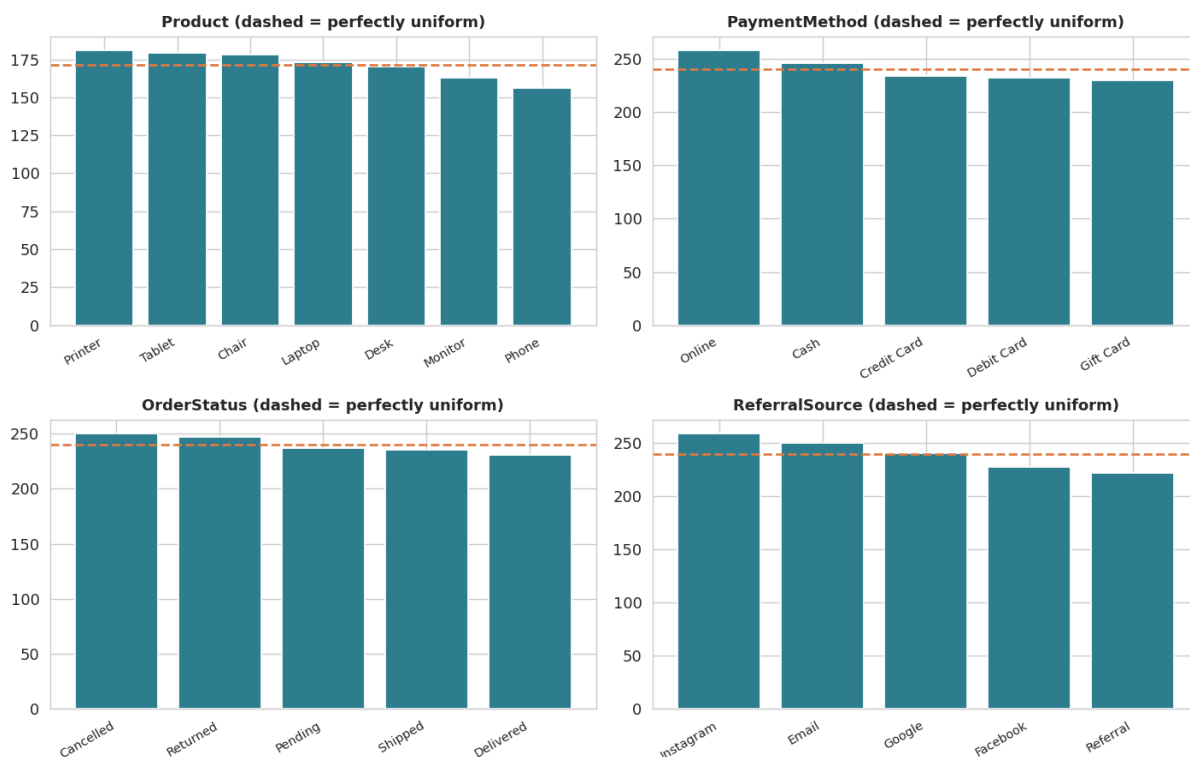


Figure 6 — Category frequencies against a perfectly uniform reference line.

4. Reference: Descriptive Statistics

Metric	Quantity	UnitPrice	ItemsInCart	TotalPrice
Count	1,200	1,200	1,200	1,200
Mean	2.95	356.41	5.48	1,053.97
Median	3.00	364.21	5.00	823.62
Std Dev	1.41	197.18	2.28	819.86
Min	1.00	11.39	1.00	11.39
Q1 (25%)	2.00	186.06	4.00	410.52
Q3 (75%)	4.00	521.57	7.00	1,578.47
Max	5.00	699.93	10.00	3,456.40
Skew	0.03	-0.03	0.00	0.89

Outlier Detection: Method Comparison

Column	IQR Fence (upper)	IQR Outliers	Z-Score Outliers	Max z
Quantity	7.00	0	0	1.46
UnitPrice	1,024.83	0	0	1.75
ItemsInCart	11.50	0	0	1.98
TotalPrice	3,330.41	8 (0.67%)	0	2.93

5. Recommendations

1. Report median order value, not mean.

Dashboards and targets should use 823.62 as the typical order. The mean overstates it by 28% and will cause revenue forecasts to run high.

2. Do not build a seasonal campaign on this data.

The H1 spike disappears under correction. Any budget shifted toward the first half would be chasing an artifact of the collection window.

3. Investigate the 8 high-value orders for retention, don't remove them.

They are arithmetically valid max-quantity purchases of premium items — the closest thing this dataset has to VIP customers.

4. Treat the 41% cancel-and-return rate as unverified.

If it reflected reality it would be the single most urgent finding here. Its suspiciously even spread suggests otherwise; confirm against the source system before escalating.

5. Source production data before any modelling.

The uniformity across every categorical variable means this dataset cannot support segmentation, forecasting or customer analysis. It is suitable for demonstrating method, not for drawing business conclusions.

6. Limitations

The final year (2025) covers only January–June, so all year-on-year comparisons are asymmetric and are flagged wherever used. TotalPrice is arithmetically derived from Quantity and UnitPrice, so their mutual correlations are partly structural rather than behavioural. Most significantly, the near-perfect uniformity of all five categorical fields is consistent with synthetic generation, which limits the external validity of any conclusion drawn here.